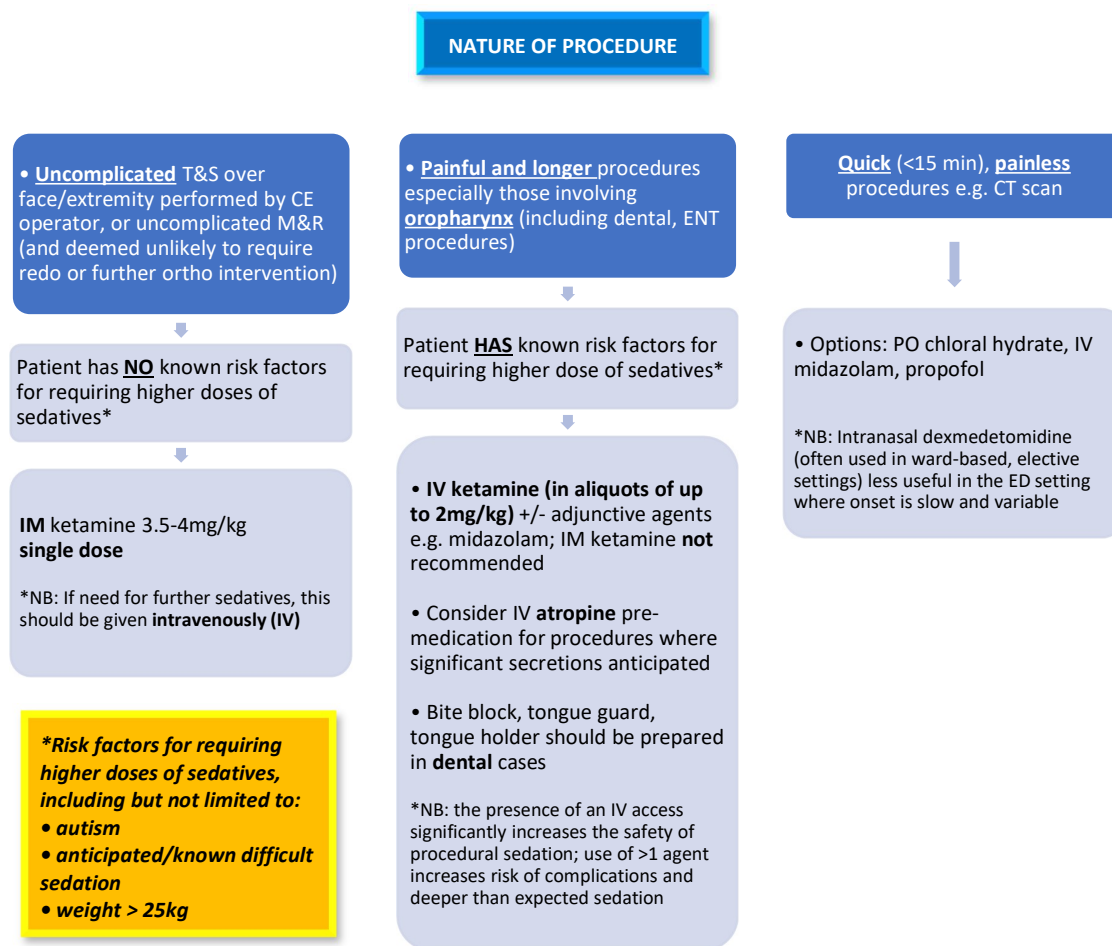


PROCEDURAL SEDATION AND ANALGESIA (PSA) IN THE CHILDREN'S EMERGENCY

Summary of guidelines:

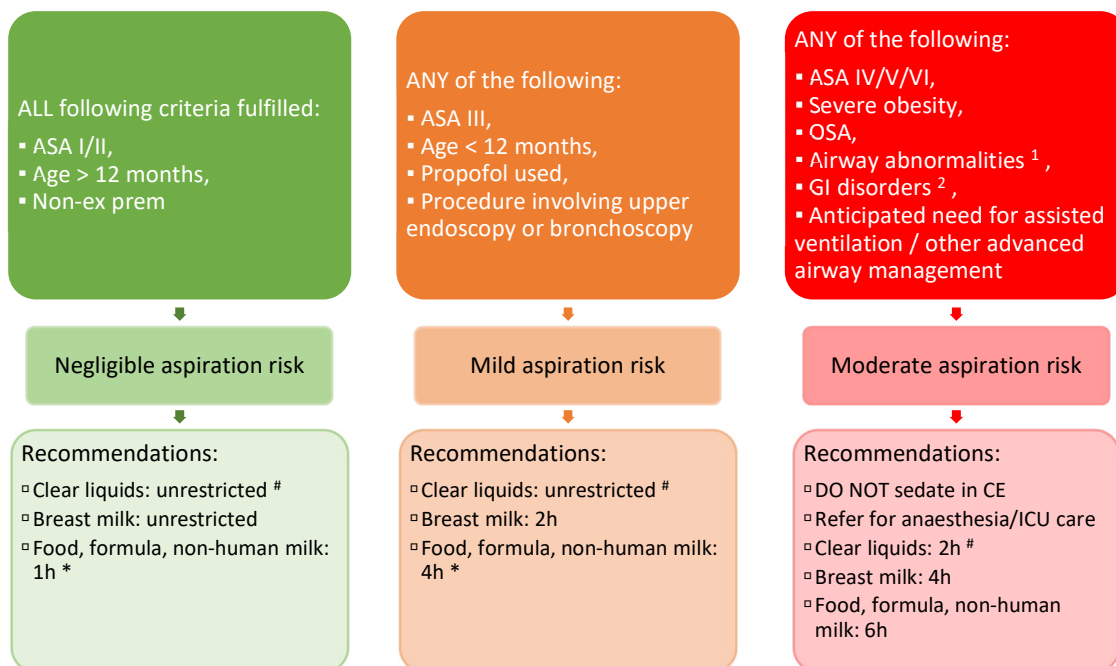
DIVISION GUIDANCE ON CHOICE OF SEDATIVE AGENTS



The use of sedative agents **must** be performed in the presence of a senior physician accredited in and familiar with sedating children and credentialed in PALS/APLS.

DIVISION RECOMMENDED PROTOCOL FOR FASTING TIMINGS

Fasting guidelines are largely based on the patient profile and the choice of sedatives used. **For ketamine-only sedation, the aspiration risk is generally negligible.**



Clear liquids are generally considered to include water, fruit juices without pulp, clear tea, black coffee and specially prepared carbohydrate-containing fluids.

* Duration is not absolute; exceptions permissible when volumes of oral intake are minor

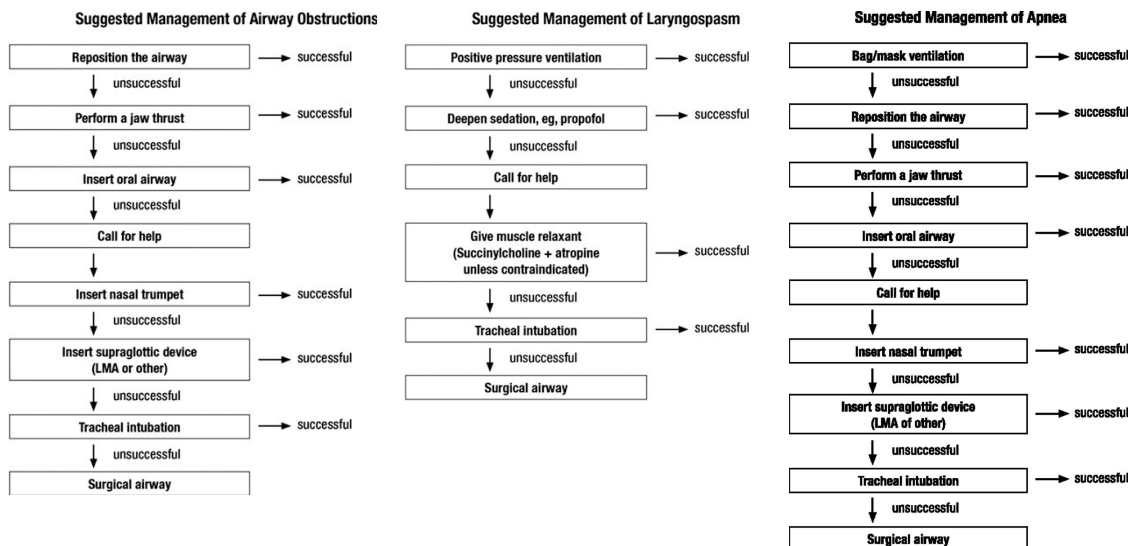
¹ e.g. micrognathia, macroglossia, laryngomalacia

² e.g. hyperemesis, oesophageal disorders, bowel obstruction (ileus, pyloric stenosis, intussusception)

MINIMUM EQUIPMENT NEEDED FOR SEDATION (SOAP-ME)

- Size-appropriate suction catheter connected to wall suction
- Face mask
- Size-appropriate bag-valve mask
- Continuous cardiac monitoring, pulse oximetry, q5min BP monitoring

TROUBLESHOOTING SEVERE COMPLICATIONS OF SEDATION



1. DEFINITION

- The practice of procedural sedation is the administration of one or more pharmacological agents to facilitate a diagnostic or therapeutic procedure while targeting a state during which airway patency, spontaneous respiration, protective airway reflexes, and hemodynamic stability are preserved, while alleviating anxiety and pain.
- Notes:
 - This includes minimal sedation, moderate sedation, dissociative sedation and deep sedation.
 - The practice of procedural sedation differs from the practice of general anaesthesia, which targets an unarousable state in which airway intervention is often required and spontaneous ventilation is frequently inadequate. [1]
- Sedation aims to reduce consciousness.
 - In contrast, anaesthesia = loss of sensation, analgesia = alleviation of pain.

2. GOALS

- The goals of sedation in the pediatric patient for diagnostic and therapeutic procedures are:
 1. To guard the patient's safety and welfare;
 2. To minimize physical discomfort and pain;
 3. To control anxiety, minimize psychological trauma, and maximize the potential for amnesia;
 4. To modify behavior and/or movement so as to allow the safe completion of the procedure;
 5. To return the patient to a state in which discharge from medical/dental supervision is safe, as determined by recognized criteria. [2]
- The aim of sedation in pediatrics differs from that in adult patients, as it is administered to control behavior and allow safe completion of the procedure. For cooperative children, non-pharmacological modalities should be considered, such as parental presence, hypnosis, distraction, and topical anesthetics, as they may reduce the need for or depth of pharmacological sedation.
- Degree of sedation can be scored using **University of Michigan Sedation Scale (UMSS)**[3, 4]:

	Degree of sedation	Cardiorespiratory effects	Adequacy of sedation
0	Awake and alert	N.A.	GENERALLY NOT DESIRED
1	Minimally sedated: tired/sleepy, appropriate response to verbal conversation and/or sound	Spontaneous ventilation adequate. Cardiovascular function maintained.	For painless, non-invasive procedures
2	Moderately sedated: somnolent/sleeping, easily aroused with light tactile stimulation or a simple verbal command	No interventions required to maintain a patent airway, and spontaneous ventilation is adequate. Cardiovascular function usually maintained.	For painful, invasive procedures
3	Deeply sedated: deep sleep, arousable only with significant physical stimulation	Ability to independently maintain ventilatory function may be impaired. Patient may require assistance in maintaining a patent airway, spontaneous ventilation may be inadequate. Cardiovascular function usually maintained.	GENERALLY NOT DESIRED

4	Unarousable (akin to general anaesthesia): unresponsive to all stimuli	Absence of airway protective reflexes, ventilatory function often impaired, and may require positive-pressure ventilation. Cardiovascular function may be impaired.	NOT DESIRED
NA	Dissociative sedation: trance-like cataleptic state characterized by profound analgesia and amnesia (specifically achieved by <i>ketamine</i>)	Retention of protective airway reflexes, spontaneous respirations, and cardiopulmonary stability.	For painful, invasive procedures

3. INDICATIONS OF PROCEDURAL SEDATION/ANALGESIA

This is a non-exhaustive list of procedures commonly performed in the Children's Emergency that may warrant procedural sedation:

Painless, non-invasive procedures; Minimally painful but anxiety-inducing procedures	Painful, invasive procedures
▪ Radiological procedures ▫ Computed tomography scans ▪ Diagnostic evaluations ▫ Eye examination in young children ▫ Naso-endoscopy e.g. foreign bodies (potentially painful)	▪ Burns debridement and dressing ▪ Orthopaedic interventions ▫ Manipulation & reduction of fractures ▫ Reduction of dislocations ▫ Wedge resection of ingrown toenails ▫ Wound inspections in young children ▪ Toilet & suture of lacerations ▪ Dental procedures ▪ Chest tube insertion

4. ALTERNATIVES AND ADJUNCTS TO SEDATION

4.1 NON-PHARMACOLOGICAL STRATEGIES

1. Patient education
 - For older patients who may benefit from mental preparation – explain step-by-step what to expect, use infographics and visual aids, involve parents to reinforce understanding.
 - For younger patients who will not be able to cope with “information overload” – explain briefly and in broad principles, avoid information overload, involve parents to allay anxiety.
2. See no needle – fear no needle
 - Prepare ‘intimidating’ equipment away from the child
 - Only expose needles after draping the child and obstructing line of sight (for face & scalp lacerations) or after the child has been adequately distracted.
3. Allow the parent or NOK to be present / go underneath the sterile sheet to distract child
4. Age-specific distraction techniques [5]
 - Young children: favourite blanket/toys, bubbles, books, audiotapes, videos/movies.
 - Older children: books, movies, virtual reality glasses, handheld video games.

4.2 ANALGESIA

- It is **important to alleviate pain** in order to relieve anxiety and distress in paediatric patients and their caregivers. **Analgesia should not be delayed** while awaiting definitive procedures under PSA.
- Intranasal fentanyl** is commonly prescribed in the CE for acute pain – refer to “Administration of Intranasal Fentanyl in the Children’s Emergency” clinical practice guideline.

4.3 LOCAL ANAESTHESIA

- Local anaesthesia for painful procedures is essential if systemic analgesics are not used. Refer to clinical practice guidelines on “Wound Management” → “Local Anaesthesia”.
- Regional and Bier’s blocks may be used for distal extremity fractures/injuries in older children. Refer to clinical practice guidelines on “Bier’s Block”, and “Wound Management” → “Regional Blocks”.

5. SEDATION

5.1 PRE-SEDATION ASSESSMENT

5.1.1 SAMPLE HISTORY

A careful risk assessment for PSA involves a good SAMPLE history, as follows:

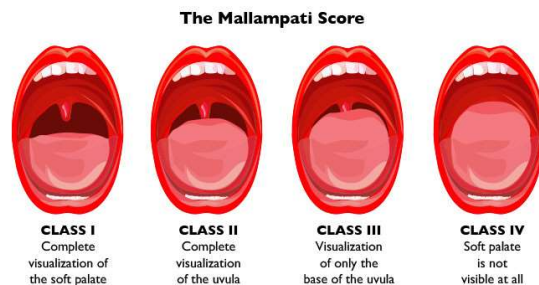
	<u>Assessed Item</u>
S	Signs and symptoms of current pathology
A	Allergy to medication, food (in particular, egg), or latex
M	Medication history, including dosage, time, route, including OTC, herbal or illicit drugs.
P	Past medical history - Comorbidities - Recent respiratory infections - Previous sedation history and associated complications - Any known physical abnormalities (including genetic syndromes) - Any known neurologic impairments that may increase potential for airway obstruction - History of snoring or OSA - History of seizure disorder - Family history particularly related to anaesthesia (e.g. malignant hyperthermia) - Review of systems: any cardiac, pulmonary, renal or hepatic dysfunction
L	Liquids and solids --- fasting time and which type of food ingested
E	Events related to the need for sedation

5.1.2 PHYSICAL EXAMINATION

A careful risk assessment for PSA also involves a focused physical examination, including the following:

- Basic parameters
 - Age and Weight
 - Baseline vital signs: HR, BP, RR, SpO2, temperature
- General examination, looking for genetic/congenital diseases associated with a difficult airway e.g. cleft lip and palate, Trisomy 21, Pierre Robin syndrome, mucopolysaccharoidosis etc.

- Focused evaluation of the airway
 - Upper airway: mouth opening, jaw size, tonsillar hypertrophy, Mallampati score



- Neck flexion: ability to pull the chin to chest (assess limitation)
 - Cardiorespiratory examination
 - Distal perfusion
- ASA stratification

Class	Health Status
I	A normal healthy patient
II	A patient with mild systemic disease
III	A patient with severe systemic disease
IV	A patient with severe systemic disease that is a constant threat to life
V	A moribund patient who is not expected to survive without the operation
VI	A patient declared brain-dead whose organs are being removed for donor purposes

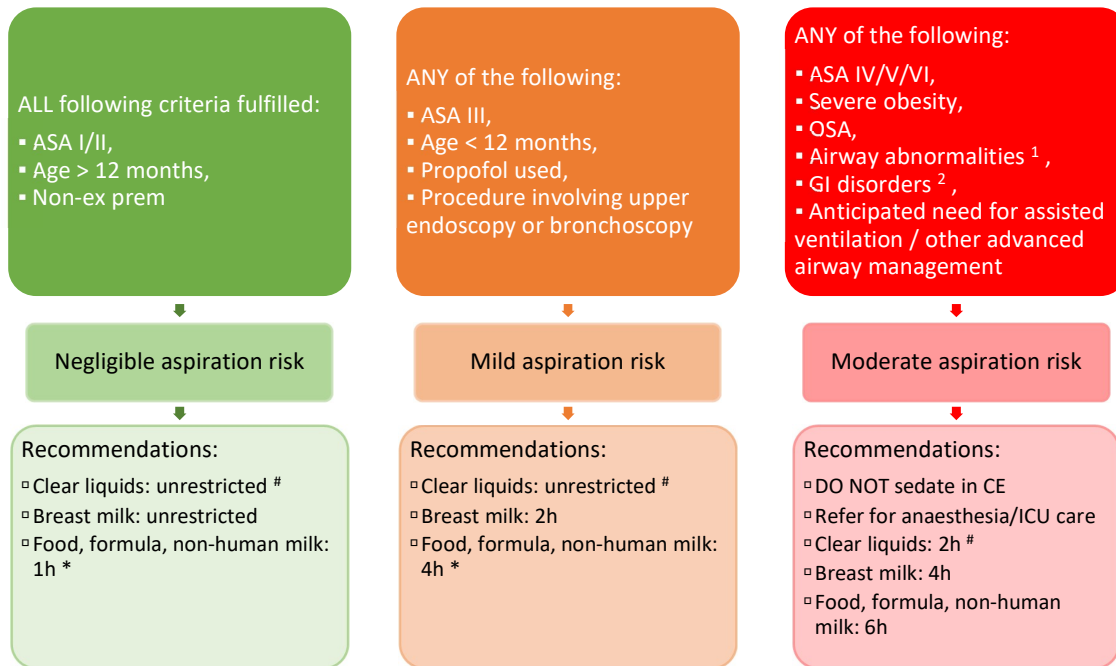
- Patients who are in ASA classes I and II are frequently considered appropriate candidates for minimal, moderate, or deep sedation. Children in ASA classes III and IV, children with special needs, and those with anatomic airway abnormalities or moderate to severe tonsillar hypertrophy require additional and individual consideration.

5.1.3 FASTING GUIDELINES FOR PROCEDURAL SEDATION [6-8]

- Despite variable fasting periods, aspiration associated with procedural sedation is rare, with the best available point estimate of its incidence 1:13,914 in a pediatric study. There are only 9 reports of aspiration-associated deaths in the post-1984 medical literature, of which 8 were during upper endoscopy in adults with ASA III or greater. None occurred in children or in healthy adults.
- The risk of aspiration during procedural sedation appears to be well under 50% of general anesthesia, and risk of aspiration-related death from procedural sedation appears to be substantially less than that with general anesthesia. Sedation for emergency procedures is not an identified risk factor for aspiration—unlike emergency anesthesia.
- There is no observed association between aspiration and non-compliance with typical fasting guidelines, either for general anesthesia or for procedural sedation, and in either adults or children.
 - Imaging has frequently identified gastric fluids and solids in pre-operative patients compliant with existing fasting guidelines, indicating that GA is regularly and widely performed in patients whose stomachs are not empty.
 - Liberalized pre-GA fasting is not associated with increase in aspiration frequency.
 - Multiple settings outside of OT which have been regularly non-compliant with typical fasting guidelines for decades, show no evidence of increased aspiration risk.

- Regurgitated clear liquids appear to represent little or no risk of aspiration morbidity.
- Three European societies of paediatric anaesthesia recently issued a joint consensus statement, reducing the NBM for clear fluids to 1 h for elective general anaesthetics in healthy children without aspiration risk factors.[9]
- Specific agents:
 - **Propofol** is the most common sedative associated with aspiration during procedural sedation[10] despite its antiemetic effects. It is unclear whether this is a result of the more widespread application of this agent, the use of concomitant opioids or the targeting of deep sedation and the potential for rapid overshoot in sedation depth.
 - **Ketamine**, unlike other sedatives, helps *preserve protective airway reflexes* [11]. There were no reports of aspiration (despite its association with vomiting and, particularly with gastrointestinal endoscopy, laryngospasm) in patients receiving this agent alone except in compromised neonates [12].
- Given the lower frequency of aspiration and mortality as compared to GA and the theoretical basis for lesser risk, fasting strategies in procedural sedation can reasonably be less restrictive [4, 12]. Since 2014, American College of Emergency Physicians (ACEP) recommend that procedural sedation in the emergency setting in both adults and paediatrics **not** be delayed based on fasting time [4].

DIVISION RECOMMENDED PROTOCOL FOR FASTING TIMINGS



Clear liquids are generally considered to include water, fruit juices without pulp, clear tea, black coffee and specially prepared carbohydrate-containing fluids.

* Duration is not absolute; exceptions permissible when volumes of oral intake are minor

¹ e.g. micrognathia, macroglossia, laryngomalacia

² e.g. hyperemesis, oesophageal disorders, bowel obstruction (ileus, pyloric stenosis, intussusception)

(Modified and adapted from Green, S M et al. "An international multidisciplinary consensus statement on fasting before procedural sedation in adults and children." *Anaesthesia* vol. 75,3 (2020): 374-385. [12])

5.2 PREPARATION FOR SEDATION

5.2.1 STAFF

- There must be at least 2 individuals (1 operator, usually the primary doctor, and 1 monitoring personnel, usually the nurse) present with the patient throughout the procedure.
- Patients receiving deep sedation should have a person skilled in establishing vascular access in paediatric patients immediately available.

5.2.2 EQUIPMENT

- Equipment needed for sedation can be recalled easily using the mnemonic **SOAP-ME** [2]:

Suction	▪ Size-appropriate suction catheter ▪ Functioning suction apparatus (eg, Yankauer-type suction)
Oxygen	▪ Adequate O2 supply and functioning flow meters/other delivery devices: ▫ Nasal prongs ▫ Face mask ▫ Non-rebreather mask
Airway equipment	▪ Size-appropriate Airway equipment (i.e. bag-valve-mask) ▪ Other airway adjuncts: ▫ Nasopharyngeal and oropharyngeal airways ▪ Advanced airway equipment ▫ LMA, laryngoscope blades (checked and functioning), endotracheal tubes, stylets
Pharmacy	▪ Pharmacy: all the basic drugs needed to support life during an emergency, including antagonists as indicated *
Monitors	▪ Monitors: continuous cardiac monitoring, pulse oximetry, q5min BP monitoring
Equipment	▪ Special Equipment or drugs for a particular case ▪ Optional: capnography/EtCO2 monitor ^

^ **Capnography monitoring:** Although the routine use of capnography appears to decrease the incidence of hypoxia and respiratory events (Level B recommendation), currently there is a lack of evidence that capnography reduces the incidence of serious adverse events during procedural sedation and analgesia such as neurologic injury caused by hypoxia, aspiration, or death. There is a lack of convincing evidence that the addition of capnography to standard monitoring in ED PSA reduces the rate of clinically significant adverse events. [4, 13]

* **Ondansetron as an adjunct:** there is some but not convincing evidence that routine use of ondansetron may reduce the incidence of sedation-related vomiting in children[14-17]. Our division currently **recommends** the **routine use of PO ondansetron** to mitigate sedation-related vomiting in CE, until further evidence advises us further.

5.2.3 DOCUMENTATION

- The pre-sedation assessment form shall be filled in by the attending doctor before the start of procedural sedation, and communicated to the attending nurse.
- The physician shall document the name, route, site, time of administration, and dosage of all drugs administered in eIMR immediately after sedation.

5.2.4 PATIENT/PARENT EDUCATION

- The attending doctor is responsible for obtaining informed consent for procedural sedation from the accompanying parent/NOK before the start of the procedure, including giving him/her the Procedural Sedation pamphlet and infographic.
- Parents must also be counselled by the attending doctor of the extra cost of \$90 (before GST) for procedural sedation in the CE.

5.3 SEDATION

5.3.1 SEDATION AGENTS

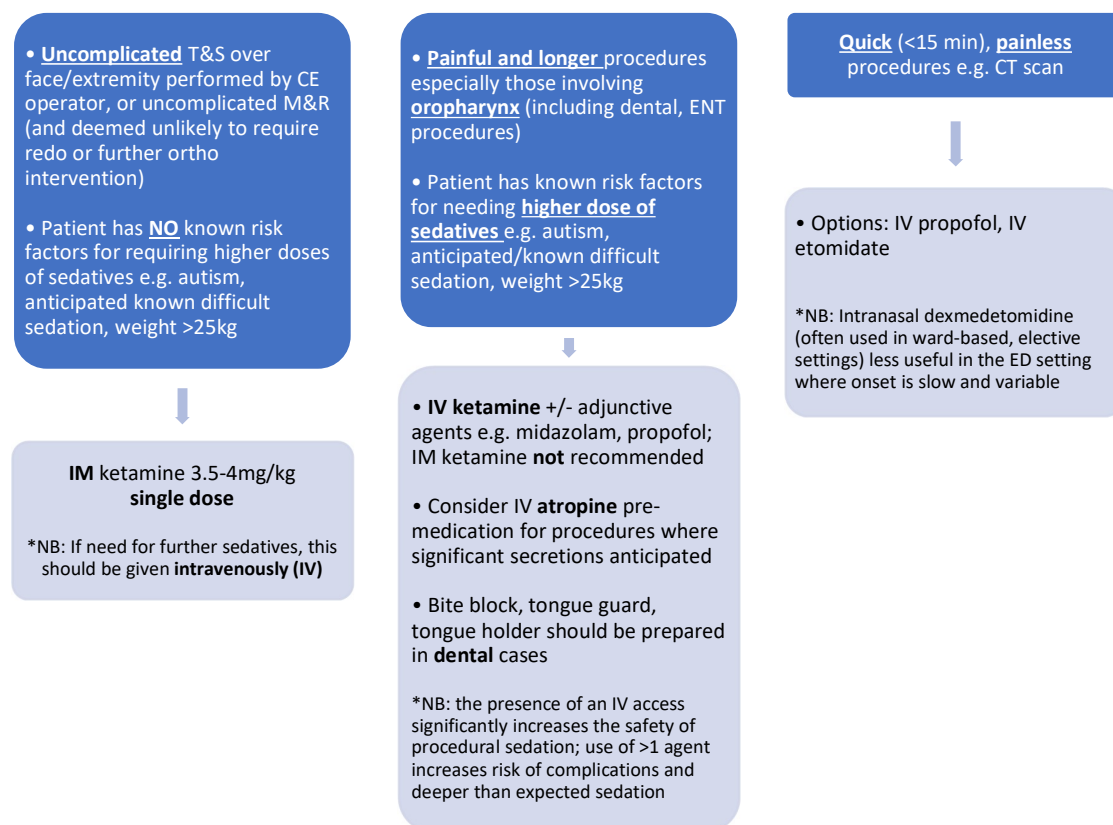
There are several agents in the emergency physician's arsenal of drugs for PSA in children. There is no one perfect drug or route of administration as the indications for PSA are varied for different procedures, age groups, pain and anxiety levels. The use of sedative agents must be performed in the presence of a senior physician accredited in sedating children and credentialed in PALS/APLS.

While top-up sedation may be required for some patients, beyond 2 top-ups, the need for continued sedation should be reviewed and expert opinion should be consulted.

Our current institution practice recommends that **IV ketamine** remain the mainstay sedative agent for most brief and painful procedures.

The following guide supports decision-making in the choice of sedative agent, with the overarching principle of aiming for the minimum lowest effective dose of sedation with a rapid onset and duration of action that minimizes the risk of complications.

DIVISION GUIDANCE ON CHOICE OF SEDATIVE AGENTS

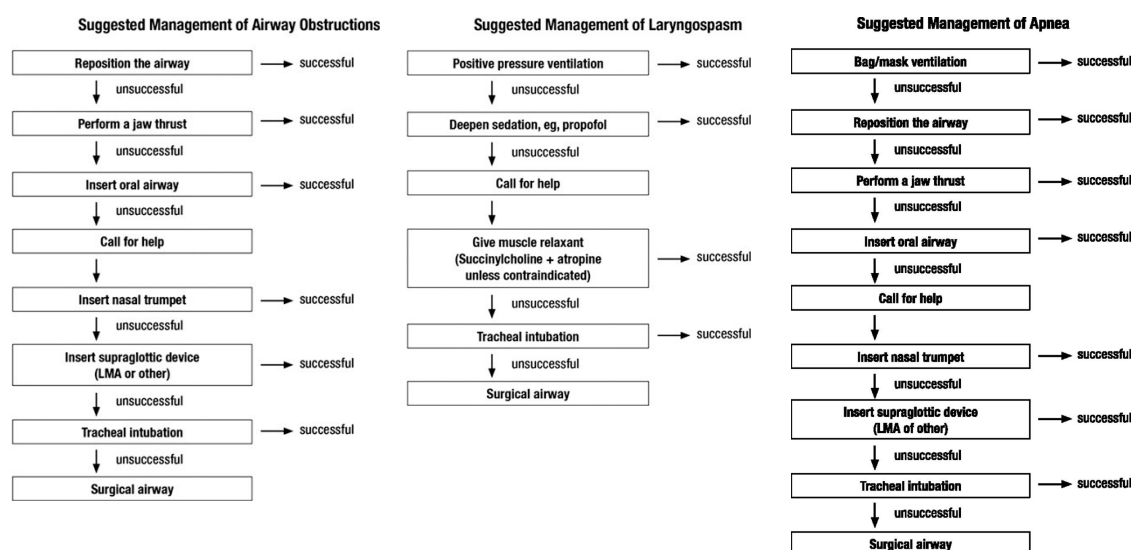


The use of sedative agents **must** be performed in the presence of a senior physician accredited in sedating children and credentialed in PALS/APLS.

[Annex A](#) outlines a non-exhaustive glossary of sedative agents available in the Children's Emergency.

5.3.2 SEDATION MONITORING AND TROUBLESHOOTING

- The most common serious complications of sedation involve **compromise of the airway or depressed respirations** resulting in airway obstruction, hypoventilation, laryngospasm, hypoxemia, and apnoea.
- Hypotension and cardiopulmonary arrest may occur, usually from inadequate recognition and treatment of respiratory compromise.
- Regardless of the intended level of sedation or route of drug administration, the sedation of a pediatric patient represents a continuum and may result in respiratory depression, laryngospasm, impaired airway patency, apnea, loss of the patient's protective airway reflexes, and cardiovascular instability[2]. **It is common for children to pass from the intended level of sedation to a deeper, unintended level of sedation** [18, 19], making the concept of rescue essential to safe sedation.
- Competency with emergency airway management procedure algorithms is therefore fundamental for safe sedation practice and successful patient rescue.



- Monitoring
 - The child's head position should be continuously assessed to ensure airway patency
 - Continuous monitoring of the following by the attending doctor:
 - Oxygen saturation and heart rate
 - Head position should be continuously assessed to ensure airway patency
 - Monitoring of adequate ventilation
 - The following parameters should be recorded by the attending nurse every 5 minutes:
 - Heart rate, respiratory rate, blood pressure, oxygen saturation,
 - For uncooperative children, the stimulation of blood pressure cuff inflation may cause arousal or agitation; in such cases, blood pressure monitoring may be counterproductive and may be documented at less frequent intervals (eg, 10–15 minutes, assuming the patient remains stable, well oxygenated, and well perfused).

5.4 POST-SEDATION

5.4.1 MONITORING

- The patient's vital signs should be recorded every 10–15 minutes.
- If the patient is not fully alert, continuous oxygen saturation and heart rate monitoring should be used until discharge criteria are met.

5.4.2 DISCHARGE CRITERIA

- Discharge criteria are as follows [2]:
 - Airway patency, ventilation, cardiovascular function, and hydration are satisfactory;
 - The patient's level of consciousness has returned to baseline;
 - The patient can sit unassisted (if age appropriate);
 - The patient is not vomiting;
 - The parent/NOK can understand discharge instructions.

5.4.3 DISCHARGE ADVICE

- Parents should be given a **post-sedation discharge pamphlet** which outlines the following:
 - Anticipated behaviour
 - Children may have short-term memory loss, sleepiness, and balance – resolves fully within 24h and thereafter they can return to normal activities
 - Feeding
 - Wait 1-2 hours after discharge from sedation to feed your child, starting with clear liquids such as juice or water in small amounts
 - Escalate appropriately if your child tolerates it without vomiting
 - Activity
 - Do not leave your child unaccompanied for the next 24 hours
 - Allow your child adequate rest
 - Do not allow your child to play sports or vigorous activities
 - Walk your child to the bathroom
 - Other procedure-related advice
 - Relevant wound care, cast care advice etc

6. ANNEX A – SEDATION AGENT GLOSSARY

Agent	Route	Dose	Pharmacology	Notes
Ketamine	IV	IV: 1-2 mg/kg (max 100mg)	IV: Onset 1 min; Duration 15 min; Recovery 60 min	- Phencyclidine (PCP) derivative, binds to NMDA receptor - Dissociative anaesthetic with sedative, analgesic, amnesia properties - Lower doses: anxiolysis, analgesia - Higher doses: dissociative sedation, amnesia and analgesia - Characteristics of ketamine dissociative state: - Dissociation: A trance-like state with the eyes open but not responding; nystagmus is typical - Catalepsy: Normal or slightly increased muscle tone, occasional muscular clonus. Adjunctive physical immobilization may be occasionally needed to control random motion - Advantages: allows maintenance of respiratory drive and airway reflexes, good cardiovascular stability (BP and HR are typically mildly increased) - Adverse effects: - Vomiting (8.4%): IM administration, high dose are risk factors - Agitation/emergence (7.6%): Age >10 yo, IM administration are risk factor - Older children should be counselled that they will dream during recovery, advised to plan pleasant dreams - Minimizing stimulation in induction and recovery may alleviate emergence reactions - Apnoea 0.8% - Laryngospasm 0.3% - Hypersalivation
	IM	IM: 3-5 mg/kg (max 100mg), after 10 minutes repeat IM 1.5-2mg/kg OR insert IV and give further doses 0.5mg/kg	IM: Onset 3-5 min; Duration 15-30min; Recovery 90-150 min	
	IN [20]	IN: 2-4 mg/kg (max 200mg) (currently insufficient evidence for routine use)	- Contraindications: - Psychosis - Intraocular trauma or glaucoma - Systemic hypertension - Active respiratory infection, active asthma increasing risk of laryngospasm *Increased ICP is not a contraindication - Reversal agent: none	

Agent	Route	Dose	Pharmacology	Notes
				Note regarding route of ketamine: RCEM now advises against IM ketamine over IV ketamine, stating whilst the use of IM ketamine is still recognised as a pragmatic option when used by a senior decision maker, clinicians should be mindful of the perceived safety benefits of having intravenous access from the start of the procedure to mitigate a rare adverse event. IM ketamine has a higher risk of emesis and a longer recovery time. IV access also facilitates repeat dosing for longer procedures. [11, 21]
Midazolam	IV IM PO	IV: 0.1 – 0.2 mg/kg (max 10mg) IM: 0.15 mg/kg (max 10mg) PO: 0.5 mg/kg (max 15 mg)	- IV: onset 2-3 min; duration 45-60 min - IM: onset 10-20 min; duration 60-120 min - PO: peak onset 30-60min; duration 2h - Antidote: Flumazenil 5mcg/kg every 60 sec to a total dose of 40 mcg/kg (adult dose 300-600mcg)	- Short-acting benzodiazepine - Provides sedation, amnesia, anxiolysis - NO analgesia - Advantages: short half-life and recovery - Specific adverse events: - Mucosal irritation when given intranasally - Paradoxical reactions in 1-15% of children (crying, combative, disorientation, restlessness)
Fentanyl	IN IV	IN: 1.5 mcg/kg (max 100 mcg) IV: 1 – 2 mcg/kg (max 100 mcg)	- IN: onset 2 min; duration 30-60 min - IV: onset 3-5 min, 30-60 min - Antidote: Naloxone 0.01 mg/kg/dose (max 2mg/dose)	- Opioid, 100x more potent than morphine - Potent analgesic, can give sedation and anxiolysis when combined with BZD - Advantages: preferred in EMD setting due to shorter onset of action, faster recovery time, tends to cause less nausea/vomiting

Agent	Route	Dose	Pharmacology	Notes
Propofol	IV	1 – 2 mg/kg; repeat 0.5 mg/kg as needed; infusion rate 100-150 mcg/kg/min	- Onset: within 1 minute - Duration: 5 – 15 min - Contraindications: Allergy to soybean oil, egg lecithin (yolk), glycerin, EDTA - No antidote	- Short-acting, lipophilic intravenous general anaesthetic - NO analgesic effect - Adverse effects: - Pain at injection site (as it is lipophilic) - Respiratory depression and apnoea (2.8 – 6.4%) - Hypotension (31%) - Longer fasting time of 4h - Advantages: rapid onset, rapid wear-off, allows close titration
Nitrous oxide (N ₂ O)/ ENTONOX®	Inhaled	Delivered via Entanox delivery system (50% N₂O with 50% O₂) Maximum 45 minutes duration for procedural sedation	- Onset: 30-60 seconds - Peak: 2-5 minutes - Duration: 2-5 minutes upon stopping inhalation - Psychometric recovery in 20 minutes - Criteria for use: - Age > 2 years old - Can tolerate tightly sealed mask to face - Cooperative with instructions - Contraindications: - Risk of expansion of air-filled closed space e.g. pneumothorax, cystic lung disease - Head injury - Age < 2 years old	- Self-administered inhalational gas for rapid onset and short-acting - Anesthesia (dissociative, euphoria, light sedation), - Anxiolysis (limited), - Analgesia (mild-moderate), and - Amnesia (mild-moderate). - Advantages: rapid onset, fast-acting, minimal cardiorespiratory effects when used as single agent; no need for NBM when used alone - Good for procedures such as: - Large joint dislocations e.g. patella, shoulder, phalanges; without significant soft tissue tears - Non-facial laceration repairs - Removal of foreign bodies from ear/soft tissues - Fracture manipulation - Burns dressings - Other painful but quick procedures - Injection of local anaesthetic - Adverse effects: - Bitter taste, dry mouth - Dizziness or disorientation - Nausea/vomiting (exacerbated with concomitant opioid use)

Agent	Route	Dose	Pharmacology	Notes
			Reversal agent: None	- Risk of diffusion hypoxia* - N₂O toxicity: Prolonged exposure to nitrous oxide can suppress liver enzymes involved with Vitamin B12 and folate metabolism - Greenhouse gas with global warming potential *Note on diffusion hypoxia: - Nitrous oxide is 24 times more soluble in blood than nitrogen and oxygen. When nitrous oxide is discontinued, nitrous oxide diffuses out of the blood into the alveoli in large volumes. If the patient is allowed to breathe air at this time the combination of nitrous oxide and nitrogen in the alveoli reduces the alveolar PO₂. This causes diffusion hypoxia and is avoided by administering 100% oxygen for 3-5 minutes post procedure. If the patients mask is off for more than 30 seconds or after discontinuing nitrous oxide 100% oxygen must be administered.
Etomidate	IV	0.2 mg/kg (max 40 mg)	- Onset 30-60 sec; Duration 5-10 min - Peak effect: 1 min	- Imidazole class of general anaesthetic – inhibits GABA receptors - NO analgesic effect - Adverse effects - Pain at injection site - Myoclonus (dose-dependent) - Concerns of adrenal suppression in sepsis - Nausea, vomiting (similar to BZDs) - Advantages: ultra-short acting, rapid onset, rapid wear-off, cardiac-stable (least likely to cause haemodynamic compromise), decreases ICP (good for traumatic brain injury)

Agent	Route	Dose	Pharmacology	Notes
Dexmedetomidine [22]	IV IN	1 – 2 mg/kg (max 200 mg) 2 – 4 mg/kg (max 200 mg)	- IV: onset 15 min; duration 45-90 min - IN: onset 30-45 min; duration 45-90 min - No antidote	α_2-adrenoceptor agonist with sedative, anxiolytic, sympatholytic effect - NO analgesic effect - Advantages: superior to PO midazolam or PO chloral hydrate in ED setting; does not cause respiratory depression; mimics deep sleep - Good for: non-invasive diagnostic scans e.g. CT scans, MRI scans

IV: intravenous; IM: intramuscular; IN: intranasal

6.1.1 COMBINATION SEDATIVES

Combination sedatives have additive risk of adverse effects and should only be used by accredited and experienced senior personnel.

Combination agents	Route	Dose	Recommended indications
Ketamine-Propofol ('Ketofol') [23, 24]	IV	- Multiple regimens described, no single method has been conclusively favourable over the other. - Ketamine:Propofol in 1:1 dosing (median dose as low as 0.8mg/kg of each medication) [25] - Ketamine:Propofol at 0.5:1 dosing i.e. ketamine 0.5mg/kg followed by propofol 1mg/kg; if for top-up, ketamine 0.25mg/kg and propofol 0.5mg/kg as necessary [26-28]	- Advantages: potentially shorter recovery time[29], decreased incidence of nausea/vomiting, lower incidence of emergence reactions, combination allows for reduced doses of each medication → fewer adverse effects - ACEP level B recommendation for safety and efficacy in children for PSA - Good for: - Invasive procedures including M&R, laceration repairs
Midazolam-Ketamine	IV		- ACEP Level A recommendation: addition of midazolam as adjunct to ketamine does not decrease incidence of emergent reactions. - ACEP Level B recommendation: addition of midazolam decreases risk of emesis.
Ketamine-Dexmedetomidine (Ketodex)	IN	- Onset 10-15 min; DOA 35-70min - Multiple regimens described, no single method has been conclusively favourable over the other - Recommended: IN ketamine 2mg/kg + IN dexmedetomidine 2mg/kg, with top-up of 1mg/kg of each drug	- Not enough evidence for routine use - Advantages: - Needle-free form of sedation, allows for reduced total dose of each medication - Good for: - Non-invasive or minimally painful procedures e.g. T&S (after application of LET gel and LA infiltration) - Not suitable for extremely anxious children or very painful procedures such as M&R of fractures

IV: intravenous; IM: intramuscular; IN: intranasal

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